

SPARC

Demonstration



"Place the Toast on the Plate."

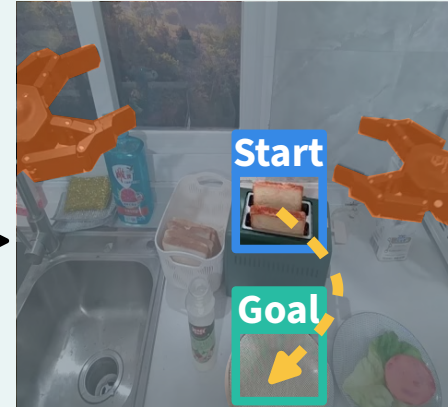
Automatic Annotation

Subtask Decomposition

Candidate Proposal + Tracking

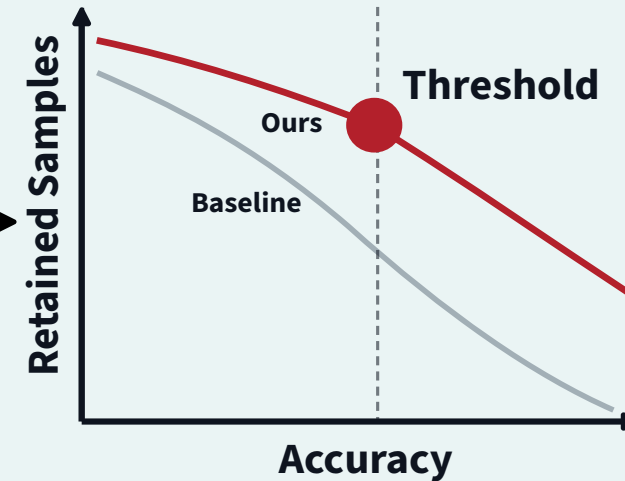
Reliability Scoring

Structured Spatial Annotations



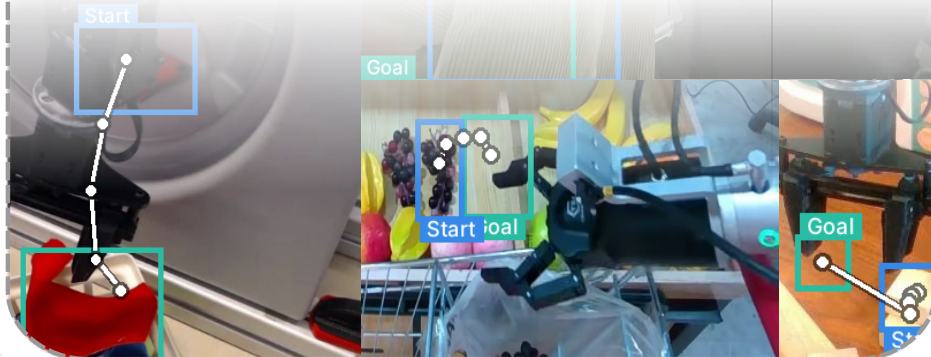
Confidence $R=0.93$

Quality Filter



Large-Scale Spatial Annotations

12 Datasets 334k Bounding Boxes
5k Object Classes 167k Robot Trajectories
225k Object Traces 42K Lang. Instructions
8k Target Locations 2k Start Locations



Grounding VQA

Vacant Space Pointing

"Locate the object the robot has to pick up to clean the whiteboard"

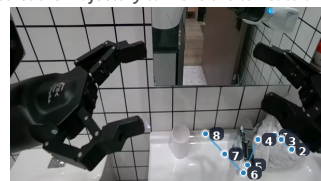


Object Localization



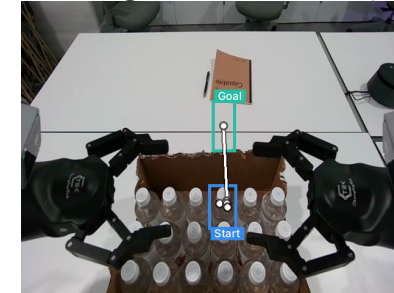
Trace Prediction

"Predict the Trajectory to move the towel to the left"



IA Bench

Human-Annotated Grounding of Interacted Objects



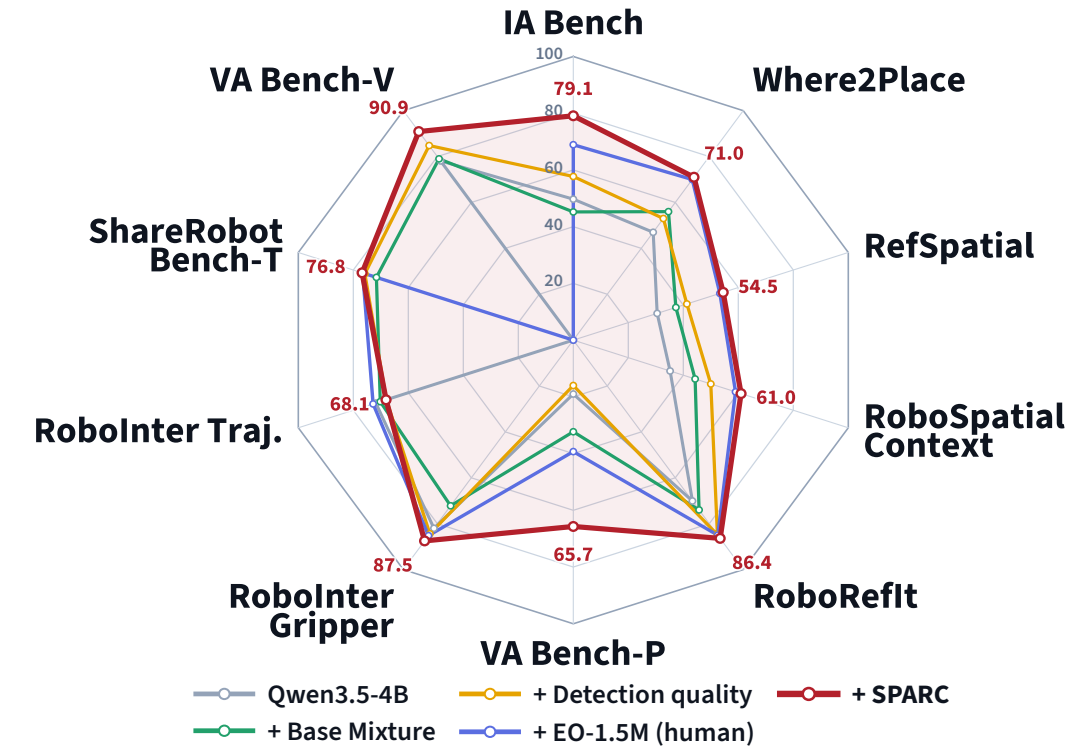
How good can models ground where an interaction happened in a robot demonstration?

1,748 Trajectories

12 Data Sources

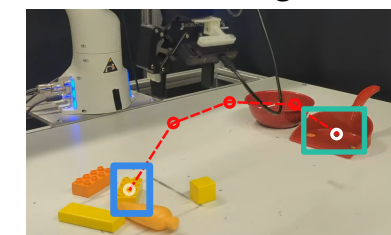
Single-Arm | Bimanual | Diverse Camera Views

Improved Embodied Spatial Reasoning

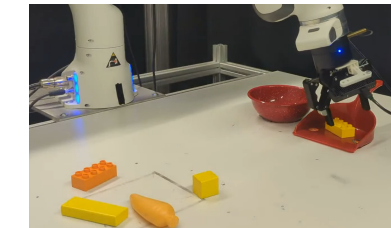


"Place the yellow lego in the red bowl!"

Reasoning



Action Generation



Better Grounding for VLAs Through Reasoning

